Text watermark detection under edits

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Research Problem and Motivation Language model watermarking is a mainstream statistical problem concerned with schemes to imperceptibly alter language-model text so that it may be uniquely identified as artificially generated, with robustness to edits applied by resource-constrained actors.

Setting. A generator samples $y_{1:n}$ from $p_t = p(\cdot \mid x, y_{<t})$ over a vocabulary $\mathcal{V}$. A watermarking scheme W for a generator consists of a key generator, a watermarked generation procedure based on a key, and a detection procedure based on the key. In green-list watermarking, the key is a sequence of independent green lists $G_t \subseteq \mathcal{V}$. Each token is in G_t with probability γ. Watermarked generation samples $y_t \sim \tilde{p}_t(v) \propto p_t(v)\, e^{\delta \mathbf{1}[v \in G_t]}$. The attacker makes at most ρn substitutions, each from a permitted set $R(y_t) \subset \mathcal{V}$ (e.g., synonyms for y_t), and is *key-blind* (i.e., the attacker knows γ, δ, p_t, not the key). After analyzing this case, we will also analyze the case when an attacker can see greenlists G_t at each position t independently with probability λ (λ-*leakage*). The detector knows the key, sees only the edited $z_{1:n}$, and must have false-positive probability at most α on any key-independent sequence. We assume edits made to fool the detector only include substitutions.

Goal. To better understand the limits of current watermarking schemes in this setting, we wish to characterize the minimax false-negative exponent $E(\rho) = \liminf_{n\to\infty} -\frac{1}{n} \log \beta_n^*(\alpha, \rho)$, which is defined in terms of the minimax false-negative probability $\beta_n^*(\alpha, \rho)$ that depends on the edit fraction ρ and false-positive limit α. After fully characterizing the exponent, we wish to obtain the critical edit fraction ρ^* which is the boundary after which $E(\rho > \rho^*) = 0$.

The Remaining Gap Green-list watermark detectors [2] are framed as hypothesis tests with the objective of minimizing false positives [3]. While prior work models human edits for asymptotic optimality [4], these approaches do not characterize worst-case attacks under fixed budgets or partial key knowledge. In practice, substitution attacks [5] and watermark stealing [6] can remove watermarks by partially learning the key. Despite their empirical success, these attack evaluations lack theoretical characterizations of the minimax false-negative probability $E(\rho)$ and critical edit fraction ρ^*. Golowich and Moitra [1] provide a watermark scheme that is undetectable, sound (low false-positive rate), and edit-robust (low false-negative rate for a lower-bounded constant fraction of adversarial edits). However, the upper-bound on $E(\rho)$ and the ρ^* beyond which the watermark fails is not fully determined against optimal, budget-limited attackers as a function of key knowledge. We aim to address this gap for a simplified, substitution-only setting in our project.

Proposed Approach To fully characterize the minimax false-negative exponent, hereafter referred to just as "the exponent," we must obtain and match lower (achievability) and upper (converse) bounds on the exponent. For the lower bound $\underline{E}(\rho)$, we first choose a detector ϕ and find the infimum exponent of this particular detector against every admissible attack (i.e., those making up to ρ substitutions). We can do this by finding and proving a certain attack strategy $\mathcal{A}^*$ dominates all other admissible attacks, and then finding the exponent for that particular detector-attack pair $\inf_{\mathcal{A}\in\mathcal{A}_\rho} E(\phi, \mathcal{A}) = E(\phi, \mathcal{A}^*)$. Then for the upper-bound $\overline{E}(\rho)$, we find the exponent for the best possible detector against that optimal attack $\sup_{\phi'} E(\phi', \mathcal{A}^*) = \overline{E}(\rho)$. To conclude these two bounds match for every ρ, we can use the fact that $\mathcal{A}^*$ is proven to be the worst attack against ϕ and then show the original ϕ achieves the optimal false-negative rate against this attack. With these matched bounds, it should be possible to derive the critical edit fraction at which $E(\rho) = 0$.

Can AI Solve this Problem We queried OpenAI GPT-6 Astra with a well-specified version of our proposed problem: characterizing $E(\rho)$ and ρ^* for green-list watermark detection under under adversarial action substitutions as described in our first section. Its response can be found here. It claimed to achieve a "substantial partial solution," with the simplifying assumption of token generation being independent. For the autoregressive version, it also obtained a lower bound for the exponent, but no upper bound. We suspect this lower-bound in the auto-regressive case is based on the result in [1]. Discussion with Claude Opus 5.5 yielded the strategy described in our proposed approach. Specifically, it recommended obtaining the lower bound for the exponent based on choosing the "green-count test in the high-entropy regime" for the detector, choosing the "priority-rule attack" as the dominant attack, and deriving the exponent for this pair. Then, the upper bound would be shown by proving the Neyman–Pearson test is an optimal detector that does not beat the first derived exponent by Stein's lemma. We consider these both to be useful starting points and indicators that the remaining upper-bound on the exponent in the autoregressive case is a non-trivial contribution.

References

[1] Noah Golowich and Ankur Moitra. "Edit Distance Robust Watermarks via Indexing Pseudorandom Codes". In: *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 2024. URL: https://openreview.net/forum?id=FZ45kf5pIA.

[2] John Kirchenbauer et al. "A Watermark for Large Language Models". In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 17061–17084. URL: https://proceedings.mlr.press/v202/kirchenbauer23a.html.

[3] Xiang Li et al. "A statistical framework of watermarks for large language models: Pivot, detection efficiency and optimal rules". In: *The Annals of Statistics* 53.1 (2025), pp. 322–351. DOI: 10.1214/24-AOS2468. URL: https://doi.org/10.1214/24-AOS2468.

[4] Xiang Li et al. "Robust detection of watermarks for large language models under human edits". In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 88.2 (2026), pp. 491–515. arXiv: 2411.13868. URL: https://academic.oup.com/jrsssb/article-abstract/88/2/491/8261309.

[5] Qilong Wu and Varun Chandrasekaran. "Bypassing LLM Watermarks with Color-Aware Substitutions". In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024, pp. 8549–8581. DOI: 10.18653/v1/2024.acl-long.464. URL: https://aclanthology.org/2024.acl-long.464/.

[6] Zhaoxi Zhang et al. "Stealing Watermarks of Large Language Models via Mixed Integer Programming". In: *2024 Annual Computer Security Applications Conference (ACSAC)*. IEEE, 2024, pp. 46–60. DOI: 10.1109/ACSAC63791.2024.00021. URL: https://doi.org/10.1109/ACSAC63791.2024.00021.